

Human–AI Collaboration in Decision Making

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Abstract— The use of Artificial Intelligence (AI) has become more common in the past several years and is being used by many different industries to help make better decisions such as in health care, finance, and business. AI systems are very fast at analysing a lot of data and can find patterns in data quickly that may be missed by humans when analysing data manually. However, while AI systems offer advantages, they have some limitations as well. AI systems may have bias in their training data, they do not have ethical reasoning capabilities, and they do not take responsibility for the outcomes of their decisions.

Human decision-makers possess strengths that AI lacks, such as understanding context, ethics, and taking responsibility for their actions. On the other hand, humans can also have inconsistencies in decision-making due to cognitive bias or their inability to make decisions with large amounts of data. Therefore, there have been a number of efforts on the part of both humans and AI to collaborate with one another to create what is now referred to as Human-AI Collaboration or Hybrid Intelligence.

Within the context of structured decision-making, human-AI collaboration was examined during this study. A decision tree classification model was trained using past human decision data to create an experimental framework based on Python programming languages. Predictive performance of the model was compared with actual human decision performance using traditional accuracy measures. The use of a loan approval process for evaluating the performance of this model as an example will demonstrate what type of structures would exist in practice. The resulting accuracy of the model was high, indicating that it can effectively imitate human decision-making, in addition to providing a fair measure of prediction accuracy.

The study concludes that there is much value in merging the predictive ability of an AI-based model with the personal judgment of a human decision-maker, but it is required that humans still be involved in the process of final decision-making to ensure ethical standards and accountability for the decisions made. The results validate the use of interpretable machine learning methods for developing transparent and accountable Human-AI collaboration systems.

Keywords— *Human-AI Collaboration; Hybrid Intelligence; Decision Making; Decision Tree Classifier; Explainable AI; Machine Learning; Trust in AI; Algorithmic Bias*

1. INTRODUCTION

Artificial Intelligence (AI) has shifted from being a concept exclusively used by academics to being a tool of general use in many different fields over the past ten years. AI systems are now commonly found in banks for fraud detection; in healthcare for diagnosing patients; and in the business world for predicting customer demand and optimizing processes. Each application of AI represents a way in which AI allows faster and more consistent decision-making compared to traditional human methods of making decisions. As these systems have become more capable, they are playing larger roles in making critical decisions.

However, as there is an increasing level of dependence on AI, these systems have also created increasing levels of concern with respect to accountability, fairness, and transparency. With respect to accountability, questions regarding who is liable for an incorrect decision made

by a given AI system and whether or not the output generated by that system can be trusted have become increasingly important. Because many AI systems work in ways that are difficult to interpret, the internal reasoning behind the decision made by an AI system can, at times, also be difficult to understand. As such, many academics and practitioners agree that AI should not be used independently in high-risk environments but rather should be used with human input and in a structured and controlled manner resulting in the formation of Human–AI Collaboration.

1.1. Human–AI Collaboration

Human–AI collaboration can be defined as a partnership between AI systems and human experts that work together to make decisions. An example of this type of collaboration might involve an AI system collecting and processing large amounts of data; identifying trends within the data; and recommending courses of action based upon those trends. A human user would then interpret the output from the AI system; apply their own knowledge and understanding of the situation; and either make their own final decision or confirm a decision made by the AI system. Neither the AI system or the human user acts in isolation; rather, each brings something unique to the table and the results are mutually beneficial to both parties.

Dolgikh & Mulesa (2021) explain that AI systems are very good at completing tasks where speed, reliability and the ability to work without creating emotional distractions are essential [1]. In contrast, humans are ideally suited to handle unforeseen circumstances and making ethical decisions through their ability to consider things that aren't represented in the training data of the AI system. When these capabilities are leveraged in an optimal way to improve the decision-making quality of both AI systems and human users will be greater than either party could have achieved alone.

1.2. The Importance of Trust in Human-AI Systems

The lack of trust can prevent users from utilizing the information produced by Artificial Intelligence and hence decrease their ability to be utilized. On the other hand, high levels of trust can lead to users relying on the output of AI-based systems without critically reviewing them, even when mistakes or errors are present.

In a study involving 337 participants, Eisbach, Langer, and Hertel (2023) found that providing users with information regarding the confidence level associated with AI prediction will aid them in making better

decisions [2]. In situations where the level of confidence expressed in AI predictions was high, users were more likely to use the output of AI. Conversely, when the level of confidence expressed in AI predictions was low users were more likely to use their own reasoning instead of AI to reach their conclusions. The ability of users to switch between sources of information to support their decisions is important to facilitate collaboration between AI and humans.

1.3. The Importance of Explainability

Explainability is an important area to establish trust in AI systems. If a user understands how a decision was made, he or she can assess the merit of that decision more effectively. XAI concentrates on making results produced by AI systems more transparent and interpretable to create a more clearly defined decision-making process by AI systems.

This study used a Decision Tree model because of its inherently interpretable format (each decision can be broken down into a series of logical conditions, allowing the user to track the decision-making process). The clarity of the decision process allows users to verify the way the decision was reached, as well as the compliance of the output with the decision criteria expected by the user.

1.4. Area of Research

Existing research mainly focuses on increasing technical capabilities in AI technology while there has been less focus on how AI will fit into human decisions when applied in real-world situations. So far, the majority of previous research has reviewed AI systems outside of human use or has proposed ways to use the AI within humans without providing actual examples of this happening.

It has been determined that researchers need to create a solution that provides the user with accurate results but also allows the user to retain an active role in the process by offering insight and understanding into the AI process. The purpose of this study is to identify and address the limitations outlined above through the development and evaluation of a collaborative Human-AI framework, which will utilize an interpretable Decision Tree model in conjunction with the loan approval process.

1.5. Scope and Aim of This Study

The objective of this research will be to design and evaluate a method to combine human judgment with AI predictions through a collaborative decision-making

framework. Additionally, a Decision Tree classifier will be developed based on a dataset of past loan approvals in order to see if it can produce a comparable representation of previous loan approval patterns.

Along with assessing model performance through use cases, wider perspective on the Human-AI system through the evaluation of the Human-AI partnership, including trust, transparency and human oversight, is evaluated. The overall goal is not to replace human decision makers but rather to show how AI can act as a tool to assist in making better quality decisions in an interpretable and ethically based manner. In summary, the findings support the concept of using Hybrid Intelligence as an effective way of deploying AI in actual practice to support decision making.

2. LITERATURE REVIEW

2.1. Development of Artificial Intelligence (AI) for Decision-Making

Artificial Intelligence has become very important in the decision-making process of business or organization due to the large amount of data being produced by digital technology. Traditional methods of making decisions before the advent of AI could no longer process in sufficient time and as a result, organizations have begun using AI systems, which process and provide valuable insights relatively quickly [6],[15]. The ability that AI has to identify patterns in large amounts of complex data allows it to provide support for superior decision-making; however, it cannot replace humans as humans provide contextual input, ethical determinations, and creativeness [7]. Therefore, the collaboration between humans and AI has evolved into a situation where AI helps improve the speed at which humans analyse information and make decisions while at the same time increasing the efficiency of humans' efforts in conducting the analysis and executing the decisions made [1],[3]

2.2. AI Application across Multiple Sectors

Artificial Intelligence is used by a multitude of organizations within healthcare, business, finance and education. In Healthcare, AI is being utilized for the diagnosis of disease, the analysis of medical imaging and the prediction of risk, enabling quicker and better-informed decisions to be made [10]. In the Business and Operational realm, AI is working to improve business functions by analysing customer behaviour, forecasting market conditions, and helping automate routine activities, leading to improved long-term strategic

planning [6]. In terms of Education, AI is providing additional functionality to educational delivery systems through machine learning, predictive analytics, and data mining of student data to identify trends and support better decision-making [8],[14]. Overall, AI enables various types of organizations across numerous fields to make more intelligent and therefore more successful decisions.

2.3. AI in Financial Institutions

Artificial intelligence (AI), and its related technologies, have revolutionized the financial services industry. Companies are innovatively using these technologies to improve their operational efficiency and accuracy [6]. AI capabilities can assist with faster and more dependable decision-making by quickly analysing financial data, analysing and identifying trends in the market, and processing client data.

The application of AI in the finance industry includes such activities as performing transaction analysis and identifying fraud [9], estimating credit risk through machine learning [12], and algorithmic trading to react quickly to changes in the market. As a whole, AI contributes to improving the speed, accuracy, and quality of decision-making in finance.

2.4. Productivity Gains through Artificial Intelligence

By automating dull tasks like data entry and document processing, AI can dramatically increase the productivity of an organisation while also providing time-saving capabilities and error reduction [5]. This frees up your employees to do higher-level or more complex work that requires analytical abilities or creative ability. In addition, AI decision support systems are continuously updated with the most recent available data in order to provide organisations with real-time insight into issues which may arise and provide organisations with the ability to react to rapidly-changing business conditions by making timely decisions [6]. In summary, integrating AI into your organisation will provide you with increased productivity through efficiency and improved quality of your decision-making!

2.5. Human and AI Collaboration

Human and AI can, together, generate options (or possibilities), which allow for better choices in ambiguous situations. Humans will have the ability to apply prior experience and knowledge of the situation to interpret the results generated by AI using extraordinary

volumes of data within minutes. Furthermore, as a result of this collaboration, inaccuracies will be minimized, consistency will be maximized, and ethical and social factors will have been integrated into the model during development. This will also give organizations more control over decision-making (due to human review) combined with the analytical power and benefits of AI. Collaborative decision-making between human and AI typically results in better, more accurate, and more effective decisions than decisions separate and individually made by either AI or human [2],[3].

2.6. Literature Review Summary

Artificial Intelligence (AI) is recognized as one of the important supporting factors that could allow an individual or group to make good decisions with big data systems (Hosseini; Nejat, 2020). AI performs excellent analytical tasks, but relies on people for their experience, values, and context to influence decision making from the data it has learned [7]. The benefits of Human-AI collaboration provide individuals or groups with better and/or more effective means of making decisions [1], [4]. However, there are challenges related to bias, transparency, and system integration that need to be addressed to properly implement AI systems [15]. As a result, addressing these challenges is essential for creating a responsible and successful enterprise-wide implementation of AI.

3. METHODOLOGY

3.1. Research Design

The study's research design is an experimental investigation of whether or not it is feasible to use machine learning to enhance human decision-making processes [8]. The proposed research will examine Human-AI collaboration by having the AI system analyse datasets and generate predictive insights that will supplement the decision-making made by humans [1].

With this framework, machine learning models will be employed to evaluate historical data and discover trends that may assist in making decisions [11]. However, humans will still be responsible for making the actual decisions. The intention is to take advantage of both human and machine strengths [7].

The methodologically defined research process will progress through several sequential stages:

- Collecting data
- Preparing data
- Feature engineering/preparation
- Training models
- Evaluating models
- Integrating human-AI decisions

The research process flow will demonstrate the capability of AI-based analytical systems to function as decision-support systems without sacrificing transparency or human oversight [6].

3.2. Dataset Description

As part of showing the relevancy of the Human-AI collaborative framework, an experimental case study was created that included a publicly available dataset that contains previous history of loan applications and the approval status of each. The dataset was used as an actual representation of information that would be included in a typical structured decision-making data source. The dataset was used to run an experiment to evaluate the applicability of the Human-AI collaborative framework, and not just in financial decision-making, but many other areas, including healthcare analytics, assistance in hiring decisions, and risk assessment.

The dataset used for this case study contains 614 data points with 13 different attributes. One of the attributes is the target variable, which is the end result from the loan applicant on their loan application. Each of the applied attributes would provide significant information that a decision maker requires to determine the outcome of a loan application. Key attributes in the dataset include the following:

- Gender
- Marital status
- Number of dependents
- Education level
- Self-employed or working for someone
- Loan applicant income
- Loan amount
- Loan payable terms
- Credit history
- Property location
- Loan status (target variable)

The target variable is the variable that tells if the application was approved or declined. The data can be either categorical value or numerical value and are required to go through preprocessing prior to providing the data into a machine learning algorithm.

3.3. Data Preprocessing

Data preprocessing plays an important role in machine learning workflows as it ensures the quality, consistency, and reliability of the dataset before model training [14]. Several preprocessing techniques were applied to prepare the data for analysis.

3.3.1. Handling Missing Values

There were missing data in many attributes in the dataset including loan amount, term of the loan, credit history, and whether or not self-employed. These missing values will be dealt with using statistical imputation methods. Usually, numerical attributes will be imputed by replacing missing values with the column mean and categorical attributes will use the mode to replace the missing values on that attribute. This ensures that there are no significant changes to the data distribution and that important information regarding the observations is retained.

3.3.2. Encoding Categorical Variables

Numeric input features are typically the method of choice for machine learning algorithms [11]. As the present dataset features various categorical variables, it is important to convert all categorical variables into a format that can be used by machine learning models. Categorical variables such as gender, marital status, level of education, and type of residence have been converted into numeric variables using a process called label encoding. This involves assigning numeric values to each category, allowing the machine learning models to learn and process the categorical information by utilizing these encoded values [14].

3.3.3. Feature Selection

There are a number of attributes in the dataset that do not provide any useful information for the prediction model. One example is the application ID, which does not influence the outcome of any decisions.

These attributes have been removed from the dataset so that there is less noise in the learning process and the models will be more efficient. Feature selection only identifies variables that have value with respect to accuracy in predicting outcomes and as a means of improving the quality of the prediction models [14].

3.3.4. Feature Scaling

There are many numerical attributes in the dataset (e.g., income and loan amounts) that have large ranges when looking at the actual values on an absolute scale.

Therefore, all numerical variables in the dataset were standardized using the Standard Scaler [14].

By performing feature scaling, all of the numerical variables within the dataset will be represented within comparable ranges; therefore, it can be assured that by representing one numerical variable with a much larger or smaller number than other numerical attributes the larger number will not overpower or dominate the learning process [14].

3.4. Model Development

In order to interpret the data and derive predictive information, we built classifiers using machine learning techniques. The primary model used for this study was the Logistic Regression model, which is one of the most common and widely applicable supervised learning algorithms for classifying observations into one of two categories [12]. Logistic Regression determines the probability that a given observation belongs to a specific category based on the values assigned to its attributes.

In this study, the Logistic Regression model is used to determine whether or not an application will likely receive approval, based on the various attributes provided with the application.

The Logistic Regression model was chosen for the following reasons:

- Interpreting the resulting output is straightforward [12].
- The algorithm performs exceptionally well on tidy and well-structured data tables [11].
- Moreover, the Logistic Regression algorithm has very low computational demand, and is widely accepted as one of the more commonly applied algorithms in practice [12].

Additionally, in order to verify the effectiveness of the results produced, we also developed a Random Forest classifier for comparison purposes. Random Forest is an ensemble of learning technique that merges many individual classification trees together in order to increase the level of accuracy in prediction and lower the potential for overfitting to occur during the training process of the model [13].

3.5. Training and Testing Strategy

To assess the predictive performance of the models, a training and testing set was created using the dataset. A ratio of 80 to 20 percent was used:

- 80 percent of the data was used for the training of the models
- 20 percent of the data was used for the test set.

The training dataset is the set of data that allows for the learning of the associations between the input variables and the output of a decision, while the testing dataset is used to test how well the model can predict unseen data [11].

The performance of the model is assessed by evaluating the accuracy of the prediction and the confusion matrix [14]. These two measures give a picture of how well the model is expected to predict and classify observations. Combined, they help to give an overview of the model's performance and also show the strengths and weaknesses of the model.

3.6. Human-AI Collaborative Framework

The last phase of this approach is focused on utilizing predictions made by an AI-based system (machine learning) to enhance human decision making through collaboration [1]. Machine learning generates predictive assessment based on a set of input data as part of its analysis web. Predictions are not designed to replace the decision-making capabilities of people rather they are a source of analytic support [6].

When humans receive the AI generated prediction, they will then benchmark the AI prediction against the relevant contextual inputs prior making the final decision. The intent of the review process is to ensure that the prediction generated by artificial intelligence is not simply accepted at face value but to provide a means for holding humans accountable to the decision-making process [7].

Both humans and AI systems provide complementary capabilities: AI systems collect and process data, while humans use contextual information and principles of ethics to assess and interpret the data [7].

4. EXPERIMENTAL RESULTS AND FINDINGS

4.1. Experimental Evaluation Framework

To determine whether the Human-AI collaborative modelling framework was effective, we conducted a set of experiments using the same experimental dataset as described in section [1]. The experiment was not just to create a prediction model but also to show how machine learning helps humans identify patterns from structured data [8]. In many real-life decisions, analysts, managers, and experts consider many characteristics before making their complete decision. The decision-making process can be improved through machine learning as it detects

patterns and trends within data that may not be easily recognizable through manual analysis.

4.2. Data Analysis and Observations

Prior to training any machine learning models, an exploratory data analysis was completed to get an understanding of the dataset [14]. Through this analysis, there was found to be some attributes which indicate if the decision will be positive or negative.

For example, those with stronger credit history would generally have a higher probability of receiving credit approval. There were also indicators within the dataset which represented financial situation such as income and loan amount which would also indicate the outcome of the decision [9].

Given the above points, it can be concluded that all datasets do contain useful information but some of this useful information may not be obvious. This is where machine learning can help us, by allowing us to discover patterns in the dataset that you weren't aware of before and provide this to you in a format easier to understand [8].

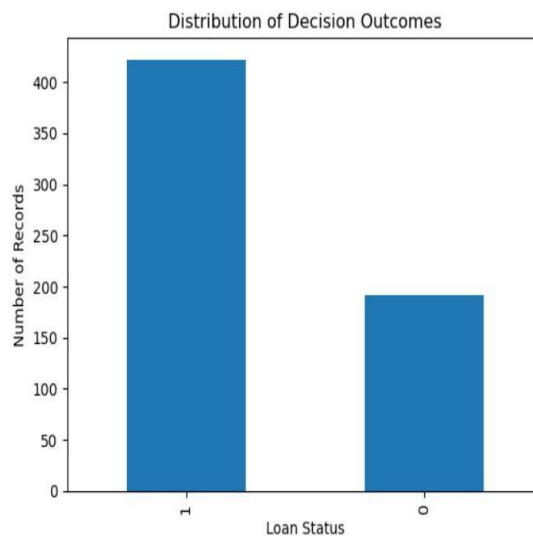


Figure 2: Distribution of decision outcomes in the dataset.

The distribution demonstrates that approved outcomes appear more frequently than rejected ones. This information helps in understanding dataset characteristics and interpreting model predictions.

4.3. Model Performance Evaluation

In order to assess the performance of the predictive models utilized in this study, classification algorithms were trained using the prepared dataset [11]. The

Logistic Regression and Random Forest classifiers were used to perform analyses on the data in order to produce predictions. Classification metrics were then used to assess the models’ performance in terms of assessing decision making within the model. The results indicate that both models produced useful patterns from the dataset (i.e., both were able to provide a model with predictive ability); therefore, both classifiers had an accuracy of 78.86% and 78.04%, respectively, on the test data; thus, Logistic Regression was rated slightly better than Random Forest for predictive ability in this experiment [12, 13].

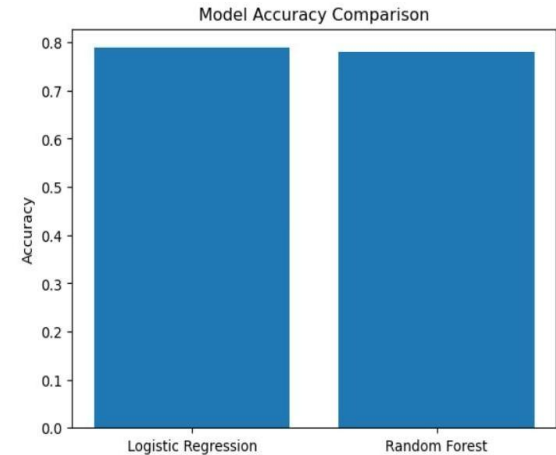


Figure 3: Accuracy comparison of machine learning models used in the experiment.

The comparison demonstrates that machine learning models can effectively analyse structured datasets and produce predictions that support human evaluation of applications. Such predictive insights can assist decision-makers in understanding patterns within the data and identifying potential outcomes.

4.3. Confusion Matrix Analysis

To further analyse the performance of the predictive model, a confusion matrix was designed. This provides an overview of the areas in which the predictive model excels and the areas in which it fails, with a clear differentiation between the two [14].

Table: Confusion Matrix Results

Actual / Predicted	Rejected	Approved
Rejected	18	25
Approved	2	79

The matrix shows that the model correctly predicted the majority of approved cases while misclassifying a

smaller number of instances. Such errors are common in predictive systems and highlight the importance of human intervention in decision-making processes [6].

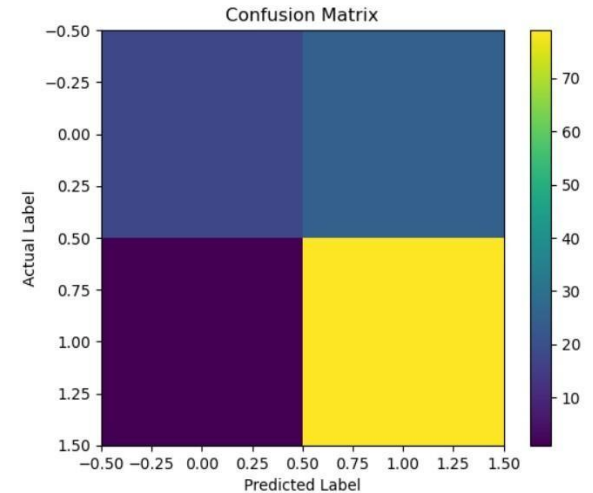


Figure 4: Confusion matrix representing classification performance of the prediction model.

In practical applications, such model outputs would be reviewed by human experts before making final decisions. This ensures that automated predictions do not replace human judgment but instead provide additional analytical support.

4.4. Implications for Human–AI Collaboration

Our studies show that there are many possible advantages of using machine learning technology within human decision-making processes [1], [2].

To illustrate, artificial intelligence can assist humans by allowing the processing of very large data sets and automatically identifying patterns which could never be seen by even the most observant of human beings. Conversely, human beings possess contextual knowledge and moral judgement, neither of which can be duplicated by any machine.

By merging human and machine strengths, human and machine working together can enhance the timeliness of decision-making, while still preserving the level of responsibility and transparency required for all parties involved. The use of a financial database within the study constitutes only one possible application of the suggested framework. This proposed framework will also apply to other diverse areas such as health care, recruitment analysis, fraud detection and risk management.

As was the case with the previous examples, while the study found reasonable prediction accuracy, the relatively small size of the experimental data set may have affected the true representation of the data's values to the real-life value in the world. Thus, in future studies, it should be appropriate to examine much larger data sets and explore other approaches to machine learning in order to validate the applicability of the proposed model of human - machine collaborative framework.

4.5. Summary of Findings

The results obtained from this study can be summarized as follows:

- Machine learning models are able to predict complex patterns and relationships in structured data [8].
- Predictive analytics provide value in supporting decision making through human data-driven decisions [6].
- AI-generated or suggested decisions can benefit from human validation prior to implementation [7].
- The combination of computational intelligence and human knowledge provides a basis for collaborating to support decision making [1].

Thus, this research illustrates how Human-AI collaborative systems can provide effective means to create intelligent decision-support systems in real-world environments.

4.6. Overall Experimental Outcome

The proposed Human - AI collaborative system has successfully been demonstrated through experimental evidence [5], [6]. Through the use of machine learning models, structured data sets have been analyzed for similarity patterns and used as an input into decision support processes.

In the case study conducted, predictive models are shown to produce a reliable classification and provide meaningful analysis of the data; however, the case study was not designed to replace human decision-makers. Instead, it was designed to demonstrate how AI-based predictions could aid humans with their evaluation of complex information.

In addition, the results of the experiment suggest that AI systems have the capability of analyzing large amounts of data and uncovering the extent to which certain variables are interconnected [8]. The predictions produced by the system increase decision-making efficiency for human users.

While the loan data set is one data set that can be used as an example of a data set used for structured decision support; the Human - AI collaborative system has many applications beyond the financial industry. Other industries, including healthcare, recruitment, fraud detection and risk assessment, have potential for development of Human - AI analytical solutions.

In conclusion, by combining machine learning with human-based expertise, the Human - AI combined decision-making process will be able to provide a higher level of transparent and informed decision-making than through singularity of machine learning or human decision-making [6], [7]. This collaborative approach ensures that machine predictions are both understood and used appropriately, while also maintaining human judgment.

5. DISCUSSION

Human-AI collaboration has been shown to improve decision-making across industries and sectors by supporting human thought processes rather than replacing them [3], [7]. In situations requiring both data and human comprehension for analysis, human-AI working relationships can complement one another [1], [4]. The study found that humans and AI work well as complements to one another. AI systems can process large quantities of data very quickly, which gives them the ability to rapidly identify patterns [8], [14]. Conversely, humans can bring their experience, ability to practically understand situations, and ability to interpret events in a meaningful way [7]. By combining the strengths of both humans and AI, the quality of decisions will improve. This is known as hybrid intelligence [1], [5]. The loan prediction model is an example of hybrid intelligence in practice. The model provides a fast and consistent way of analysing historical data to predict loan approvals. However, because humans are involved in the process, these predictions will be accurately assessed and addressed at the manual level in a timely manner to create accelerated and reliable decisions. In turn, the workload is reduced at the manual level however, the quality of the decision is still preserved [6]. Another major benefit is an increase in efficiency. AI significantly reduces the requirement for manual analysis by providing valuable insight into the decision-making process, thus allowing decision-makers to spend more time analysing and evaluating the results; rather than processing raw data [6], [7]. Consequently, decisions can be made faster, without sacrificing accuracy. This is a particularly valuable proposition in

industries such as finance where time and precision are key [9].

Collaborating with AI allows for a more data-centric way of making decisions, rather than basing them solely on limited or assumed information, as there is actual patterned data to help steer decision-making toward a more trusting outcome and a more equal process. When looking specifically at loan predictions, past data can help determine the best patterns of loan predictions for similar types of loans and help to ensure uniform outcomes [12], [13]. Another benefit that comes from this method of collaborating with AI is that it creates a continuous learning process between the user and the system, as both parties can learn from the other and continually improve based on how both have been used. An additional advantage lies in the scalability of the organisation, as by implementing AI into the organisation, you can increase productivity by being able to handle a greater number of your products or services than the current manual labour methods you are currently using. Therefore, using AI will enable you to process many more loan applications; as a result, organisations can manage larger volumes of workload and meet deadlines without quality concerns [9]. A key advantage is that confidence in decision-making is enhanced when decisions are made with the support of both AI and human judgement [2], [4]. The use of both AI analysis and human judgement leads to an increase in trust and an increased ability to justify why a decision was made. Decision-making is based on data-driven rationale from AI, while the actual real-world context of the situation is considered by humans to make sure the decision was appropriate to be made. All this creates greater confidence in the outcomes of the decision-making process [7].

The use of artificial intelligence (AI) is providing a lot of similar information to many different areas, including healthcare, education, and finance [10], [6]. AI gives insights to professionals, but it is humans who ensure that decisions made are valuable and meaningful [7]. As a result, it is clear that AI and humans working together is much more productive than working apart. In conclusion, the conversation shows that AI and humans working together can help us to make faster, more accurate, and more consistent decisions when they are used together! A great example of this collaboration is the loan prediction model developed for use in this project.

6. CONCLUSION

This study demonstrates the necessity for humans to work together with AI so that they can assist them in making daily decisions [3], [7]. As the presence of AI has begun to expand more into different sectors, researchers have been able to determine that humans and AI working together will yield the best possible outcomes [4], [6]. One of the main points to emerge from this study is that organisations must understand that AI is a support mechanism and not a replacement for humans [5], [7]. Although AI is skilled at processing data and generating predictions, humans will always need to be a part of understanding those predictions and generating final decisions based on these predictions [7]. Having human and AI work together will create more reliable and accurate outcomes. For example, the model used for loan prediction in this study is a great demonstration of how this collaboration occurs in the real world through case studies. In particular, AI is very good at processing large amounts of data and providing timely results to humans [8], [14]; thus, humans enhance the use of these results by using them properly to make decisions [7]. Having AI assist humans with decision-making enables greater efficiency and accuracy. Finally, another important takeaway from this study is that the collaboration between humans and AI provides for a much more structured and data-driven decision process [14], [15]. Decisions are no longer made based on primarily subjective reasoning but rather on actual data and analysis, resulting in greater consistency and reduced chances of error in the decision-making process. This is particularly useful in critical areas such as finance [9]. Continuously improving the collaboration between humans and AI will allow both sides to learn from one another. AI algorithms can continue to learn as they access new data, thus improving accuracy of output. Conversely, as humans provide accurate feedback into their respective systems, both systems improve over time through various learning methods [7]. As such, it represents a very solidly structured base of advancement for future decision support systems.

When viewed in a larger context natural/AI collaboration becomes a uniquely effective solution to very complex problems [1], [7]. It harnesses both the speed and precision associated with AI and the expertise and judgment associated with human beings; thus, this system is relatively adaptable and viable for an entire range of applications.

Overall, through effective collaboration between humans and AI (artificial intelligence) systems in the decision-making process, the result will be a faster, more accurate, and more dependable method of making those decisions [3], [4]. This research has demonstrated ideal knowledge-based (i.e., by merging) a human's intellectual capacity with an AI system will produce superior results from when just either is used individually [3], [4].

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